

Guidelines for Reporting About Network Data (GRAND)

The Guidelines for Reporting About Network Data (GRAND) are designed to facilitate clarity and transparency in network research reported in scientific articles. Although the information reported in a given manuscript will depend on the research question and other contextual details, these guidelines identify the minimum information that should always be reported. They provide authors with guidance about what to report in articles, and provide peer reviewers with guidance about what to expect should be reported in articles.

Notes –

- These guidelines should be used to describe the final network(s) upon which analyses are performed, after any pre-analysis transformations have been applied.
- The items should be reported somewhere in the main text of a manuscript, typically in the methods section. Alternatively, they may be summarized in a table.
- Some items apply only in specific cases, which are noted by an explicit ‘if’ statement in the guideline.
- These guidelines are designed to apply to many common types of network data, authors may need to adapt guidelines to accommodate unique cases.
- If an item is unknown, this should be reported.

1. Data: The items in this section are intended to provide a description of the network dataset as a whole.

#	Item	Reporting guideline	Notes
1.1	Source	Report whether the data are observed or synthetic	Observed data are empirically collected using direct (e.g. survey, experiment) or indirect (e.g., archival, trace) methods. Synthetic data are generated by a model or simulation.
1.2	Multiple	If the data contain multiple networks representing the same system at different temporal waves, report the number of waves. If the data contain multiple networks that each represent the same type of system, report the number of networks.	Longitudinal network data can capture the same system at multiple cross-sectional waves. For example, if a classroom's social network was observed in January, February, and March, report that the data contain 3 waves. A single dataset can contain one network from each of multiple sites, or may contain one ego network from each of multiple ego respondents. For example, if the data describe friendship networks in 10 different classrooms, or animal interactions in 10 different sites, or core discussion networks of 10 different people, report that the data contain 10 networks.
1.3	System	Report the type of system that the data are intended to represent.	A network is a representation of a connected system, for example, a university, the human brain, a food chain, or a focal individual's personal relationships.
1.4	Location	If observed (see 1.1) , report the geographic location where the data were obtained.	The reported location may be precise (e.g., Fiordland, New Zealand's Doubtful Sound; Lusseau, 2003) when it is substantively significant and there are no ethical concerns (e.g., when observing wild dolphins). Alternatively, the reported location may be more general (e.g., the United States), when the specific location is unimportant or there are ethical concerns (e.g., observing young children in a classroom).
1.5	Date	If observed (see 1.1) , report the date(s) that the data are intended to represent.	This item is intended to capture the date(s) that the data represent, which is not necessarily the same as the date(s) that the data was collected or compiled. For example, if workplace interactions were observed on 15 August 2005, then report that the data represent 15 August 2005. In contrast, if historical marriage records from 1400-1434 were compiled for analysis in 1993 (e.g., Padgett & Ansell, 1993), then report that the data represent 1400-1435.
1.6	Data	Report how and where to obtain the data, or why the data are not available.	This item follows the recommendation that "researchers should share the network data and materials necessary to reproduce reported results" (Neal et al., 2024).
1.7	Citation	<i>OPTIONAL: Provide a citation to a source where the data is more fully described, or that credits those who originally collected or generated the data.</i>	This item follows the Joint Declaration of Data Citation Principles (Martone et al., 2014).

2. Nodes: The items in this section are intended to describe the nodes in the network(s).

#	Item	Reporting guideline	Notes
2.1	Mode	Report the number of types of entities the nodes represent.	In unimodal networks, all nodes represent the same type of entity (e.g., in a unimodal social network, all nodes represent people). In multimodal networks, nodes represent more than one type of entity (e.g., in a bimodal pollination network, some nodes represent plants and other nodes represent pollinators).
2.2	Nodes	Report what the nodes are intended to represent using a word or phrase. If multimode (see 2.1) , report each type of entity the nodes are intended to represent.	For example, in a unimodal social network, report that the nodes represent people. In a bimodal pollination network, report that the nodes represent plants and pollinators.
2.3	Boundary	Report whether the set of nodes that should be included in the network was bounded by responses to a name generator question (ego network), by an <i>a priori</i> list (bounded whole network), or was not bounded (unbounded whole network).	When the nodes that should be included in the network (i.e., the alters) are bounded by a single respondent's answers to a name generator question, it is an ego network. When the nodes that should be included are bounded by an <i>a priori</i> list of all nodes in the system or setting, it is a bounded whole network. When there is no complete list of eligible nodes, and nodes are included if they meet inclusion criteria or are identified using a sampling procedure, it is an unbounded whole network.
2.4	Count	Report the number of nodes. If multiple (see 1.2) , report the mean, standard deviation, minimum, and maximum number of nodes across the waves or networks. If multimode (see 2.1) , report this item separately for each type of node.	This value is sometimes called 'network size' or 'graph order', and abbreviated $ V $. In a single unimodal network, report the number of nodes (e.g., 10). In a single multimodal network, report the size of each node set (e.g., 5 plants and 5 pollinators). In a longitudinal network dataset, summarize network size across waves, or in a sample of ego networks or networks measured in different settings, summarize network size across the networks (e.g., $M = 20$, $SD = 1$, $Min = 18$, $Max = 22$).

3. Edges: The items in this section are intended to describe the edges in the network(s).

#	Item	Reporting guideline	Notes
3.1	Plex	Report the number of types of relationships that edges are intended to represent.	In many networks, all the edges represent the same kind of relationship (e.g., in a friendship network, all the edges represent friendships). However, multiplex networks contain multiple types of edges that each represent different relationships (e.g., a network in which people could be connected by friendship and/or kinship; a network where plants could be connected to pollinators by pollination, and plots could be connected to plants by genetic similarity).
3.2	Direction	Report whether the edges are directed or undirected. If directed , report what the direction of the edge means. If multiplex (see 3.1) , report this item separately for each type of edge.	An undirected edge connects two nodes (e.g., i and j are connected), and a network composed of only undirected edges is sometimes called symmetric. A directed edge points from one node toward another (e.g., $i \rightarrow j$), and a network that contains directed edges is sometimes called asymmetric.
3.3	Weight	Report whether the edges are weighted, unweighted, or signed. If weighted , report the meaning, level of measurement, and range of weights. If multiplex (see 3.1) , report this item separately for each type of edge.	In an unweighted network, edges are either present (usually coded 1) or absent (usually coded 0). In a weighted network, present edges are assigned a weight that may represent a count, frequency, intensity, capacity, valence, etc. These weights may be measured on a continuous or ordinal scale, and may include positive or negative values. In a signed network, present edges are either positive (usually coded +1) or negative (usually coded -1).
3.4	Loops	If the network does contain loops, or can contain loops , report this.	A loop is an edge that connects a node to itself.
3.5	Edges	Report what the edges are intended to represent using a word or phrase. If signed (see 3.3) , report what the positive edges are intended to represent, and what the negative edges are intended to represent, using words or phrases. If multiplex (see 3.1) , report this item separately for each type of edge.	In a social network, the edges often represent a social relationship such as “friendship” or “kinship”. In a plant-pollinator network, the edges represent “pollination”. In a signed network, positive edges may represent “friendship”, while negative edges may represent “enmityship”.
3.6	N-partite	If multimode (see 2.1) and edges can only connect different-type nodes , report this and describe the network as N-partite.	For example, a plant-pollinator network is bipartite because edges can only connect a plant node to a pollinatory node.
3.7	Hypergraph	If a single edge can simultaneously connect more than two nodes , report this and describe the network as a hypergraph.	In a hypergraph, a single edge can simultaneously connect more than two nodes. For example, a hypergraph representing authorship connect all authors of a single paper via a single hyperedge.
3.8	Count	Report the number of edges (ignore weights, count directed edges separately). If multiplex (see 3.1) , report this item separately for each type of edge. If multiple (see 1.2) , report the mean, standard deviation, minimum, and maximum number of nodes across the waves or networks.	This value is sometimes called ‘graph size’ and abbreviated $ E $.

4. Procedure: The items in this section are intended to explain how the network data were obtained.

#	Item	Reporting guideline	Notes
4.1	Method	If observed (see 1.1), report the method used to collect the data using a word or phrase. <i>OPTIONAL: Provide a detailed description of the data collection methods, including (as applicable): how the network was bounded, how participants were recruited, the modality and procedures of data collection, and the exact wording of the question(s) used to collect data.</i> If synthetic (see 1.1), report the name of the data generating model with a citation, and the values of the function parameters. <i>OPTIONAL: Provide a detailed description of the data generating model, including its theoretical foundation and the role of each parameter.</i>	Common methods of network data collection include: survey, interview, observation, archival. Common data generating models include: Watts-Strogatz small world model, Preferential attachment model, Configuration model
4.2	Ethics	If the data or their collection could present ethical concerns, report whether institutional or ethical approval was obtained, the approving entity, and approval date. <i>OPTIONAL: Provide a detailed description of any approvals, procedures for obtaining informed consent, potential risks, and data security practices such as deidentification.</i>	Ethical concerns often arise when data are collected from or about 'human subjects,' in which case research activities are subject to approval and oversight by an external entity such as an Institutional Review Board (IRB). However, ethical concerns can also arise in non-human subjects data, for example, when the data could be used to cause harm.

5. Limitations: The items in this section are intended to describe limitations of the data collection or generation process that could introduce error, bias, or distortion.

#	Item	Reporting guideline	Notes
5.1	Degree	If a node's minimum or maximum degree is constrained by the data collection or generation design, report any restriction(s) on nodes' degree.	Degree constraints often arise when collecting data using name generator questions with 'fixed choice' or 'nomination caps' (e.g., asking respondents to report the names of exactly X, or up to X, contacts).
5.2	Nodes	If bounded whole network (see 2.3), report the percent or number of missing nodes.	Missingness occurs in a bounded network when nodes that are members of the system represented by a network are missing from the network. It is important because "because it distorts the overall configuration of actors in a system, may render an entire analysis meaningless" (Laumann, Marsden, and Prensky, 1983).
5.3	Edges	If some edges are missing because their presence or absence is unknown or was not measured, report the percent of all possible edges that are unknown or were not measured.	The lack of an edge between two nodes could represent either (a) certainty that the two nodes are not connected, or (b) uncertainty about whether or not the nodes are connected. This item is intended to capture the latter type of missingness.

6. Transformations: The items in this section are intended to describe transformations that were performed on the raw data to obtain the final analytic data.

#	Item	Reporting guideline	Notes
6.1	Symmetrizing	If directed edges were transformed into undirected edges , report the transformation method and the percent of connected dyads that had non-reciprocal edges before the transformation. <i>OPTIONAL: Provide a detailed description of the transformation method.</i>	Common transformation include confirmed edges (i.e., transforming $i \rightleftarrows j$ into $i - j$) and unconfirmed edges (i.e., transforming $i \rightarrow j$ or $i \leftarrow j$ into $i - j$).
6.2	Binarizing	If weighted edges were transformed into unweighted edges , report the transformation method and the percent of edges that were deleted as a result of the transformation. <i>OPTIONAL: Provide a detailed description of the transformation method.</i>	Common transformations include thresholds (i.e., count all edges with weight $> X$ as present) and backbone models (i.e., count edges with statistically significant weights as present; Neal, 2022).
6.3	Projecting	If the network was generated from two-mode or non-network data such that edges represent similarities, co-occurrences, or statistical associations , report the transformation method. <i>OPTIONAL: Provide a detailed description of the transformation method.</i>	Common transformations include projection (Breiger, 1974) and (weighted) correlation networks (Langfelder & Horvath, 2008).
6.4	Excluding	If any nodes have been excluded from the network , report the exclusion criteria and number of nodes that were excluded. <i>OPTIONAL: Provide a detailed rationale for the exclusion of these nodes, and the analytic implications of their exclusion.</i>	Nodes are sometimes excluded from a network if they are isolates (i.e., have no edges), or if they are members of a small component.
6.5	Aggregating	If any nodes or edges have been aggregated , report the aggregation method. <i>OPTIONAL: Provide a detailed description of the aggregation method.</i>	Individual nodes are sometimes aggregated into supernodes when they are structurally equivalent or belong to the same community. In multiplex networks, edges connecting the same dyad (e.g., i and j are connected by friendship and kinship) are sometimes aggregated into a single edge.
6.6	Imputing	If any missing nodes or edges were imputed , report the imputation method and the percent of nodes or edges in the network that were imputed. <i>OPTIONAL: Provide a detailed description of the imputation method.</i>	Link prediction methods are sometimes used to infer the location of missing edges.

7. Structure: The items in this section report basic structural properties of the network. It is intentionally brief because the relevance of most structural properties depends on the research question or analysis.

#	Item	Reporting guideline	Notes
7.1	Connected	Report whether the network is connected or unconnected. <i>OPTIONAL: If unconnected, report the number of components, fraction of nodes in the largest component, and/or number of isolate nodes.</i> If directed (see 3.2), report whether the network is strongly connected, weakly connected, or unconnected. If multiplex (see 3.1), report this item separately for each type of edge. If multiple (see 1.2), report the number of waves or networks that are connected.	In a connected network, there exists a path between every pair of nodes.
7.2	Connectivity	Report either the edge density or mean degree (ignore weights). If multimode (see 2.1), report this item separately for each type of node. If multiplex (see 3.1), report this item separately for each type of edge. If signed (see 3.3), report edge density or mean degree separately for positive and negative edges. If multiple (see 1.2), report the mean, standard deviation, minimum, and maximum edge density or mean degree across the waves or networks. If ego network (see 3.2), also report whether the edge density or mean degree was computed with or without ego. If hypergraph (see 3.7), report the mean degree for nodes and the mean size for hyperedges.	Edge density (i.e., the fraction of possible edges that are present) is more commonly reported in social sciences, while mean degree (i.e., average node degree) is more commonly reported in physical sciences. These values can be computed from the network's number of nodes (see 2.4) and number of edges (see 3.8), and therefore are redundant with information reported earlier. However, they are included in GRAND for their function as a checksum or "sanity check" (see Bagrow & Ahn, 2022).
7.3	Reciprocity	If directed (see 3.2) , report the percentage of edges that are reciprocated.	In a directed network, an edge is reciprocated if $i \rightleftarrows j$, and is non-reciprocated if $i \rightarrow j$ or $i \leftarrow j$.